Module 6 — Patent Landscape Reports

45-minute block · TIP4PATLIBS course material

> Landscape analyses after **Riccardo Priore**, Centro PATLIB, AREA Science Park. This module is contributed material, reworked to match the course's look. The analytical approach and the worked example are his.
>
> **How to read this document.** The running text addresses **you, the participant**. Boxes labelled **Trainer** are for whoever is **running the session**; boxes labelled **Trap** are traps that have caught people before.

Learning objective

I can define a patent corpus with a search strategy I am able to defend, run a battery of standard analyses over it, and assemble the result into one self-contained report a client can keep.

Prerequisites

- **Module 1** — TIP is running, PATSTAT connects.
- **Module 3** — you can read a query and judge a result.
- **Module 4** — you know that an applicant name is not a company.

Sub-objectives

By the end you can:

1. **Build a corpus as an intersection**, not as a keyword search: keywords **AND** (IPC OR CPC), and explain what each half of that rule is for.
2. **Justify an exclusion** — say why generic terms and bare acronyms were deliberately left out, and what precision that bought.
3. **Recognise the report pipeline's one shape**: a question in markdown → one code cell that reads the shared corpus, queries PATSTAT and builds one figure → one `record(...)` call.
4. **Read a landscape report critically** — which of its thirteen charts answers *who*, which answers *where*, which answers *when*, and which answers *how it connects*.
5. **Say what the assembled artifact is**: one self-contained HTML file plus one workbook, no internet required.

Material

Folder	`6_patentreports/2_antibiotic_resistance_rebuild/`

Notebooks	`1_dataset_and_search_strategy` · `2_core_landscape_analyses` · `3_advanced_analyses` · `4_assemble_report`
Artifact	`4_report/antibiotic_resistance_report.html` + `…_report_data.xlsx`
Runs on	EPO TIP, PATSTAT PROD (notebook 2's citation analysis uses BigQuery)
Ships	**pre-executed** — 29 of 29 code cells, outputs kept

> **Trap — This module ships with its outputs, and that is deliberate.** Modules 1–5 clear their outputs because participants are meant to run them. Module 6 is read as a **finished report** in a showcase. The outputs *are* the deliverable — never re-run the cells to "tidy" them.
>
> **Trainer.** The sibling folders `1_antibiotic_resistance/` (the imported reference) and `2_antibiotic_resistance_mvp/` (the frozen MVP) are **not** course material. Use the rebuild. If someone asks about t-SNE clusters or triadic families, those live in the reference folder and are a future phase of the rebuild — say so rather than improvising.

Phase 1 · Introduction (≈ 7 min)

The question this module answers

> *A regional cluster manager asks: "give us a picture of what is happening in antibiotic resistance." Where do you start?*

Everyone starts in the same place — a keyword — and this is where a landscape report is won or lost. Ask the room what they would search for. Somebody will say **"drug resistance"**. Somebody will say **MRSA**.

Both are the wrong answer, and for opposite reasons:

- **"drug resistance"** is dominated by **cancer**. Your antibiotic landscape would be an oncology landscape wearing the wrong label.
- **MRSA, VRE, ESBL** — bare acronyms — are noisy. They collide with unrelated strings and appear in documents that are not about the thing at all.

Neither problem is visible in the result. You get a large corpus, it looks healthy, and it is about something else.

The fix is not a better keyword. It is a **rule with two independent halves**: a family must match a keyword **and** carry a relevant classification. Keywords bring recall; classification brings precision. Neither alone is defensible.

	Teaching and learning activity	Time
Opening	Trainer asks for search terms for "antibiotic resistance"; collects them	2 min
Tension	Trainer names what "drug resistance" and bare acronyms actually return	3 min
Framing	Trainer states the module's rule: **keywords AND (IPC OR CPC)** — and that the exclusions are part of the method, to be written down	2 min

Phase 2 · Working through (≈ 28 min)

Four notebooks, one pipeline. **Run them in order 1** → **2** → **3** → **4** — each writes what the next reads.

Step 1 — Define the corpus (10 min) · `1_dataset_and_search_strategy.ipynb`

Eight steps that end in one file.

	What happens
1	Connect to PATSTAT PROD via the SQLAlchemy ORM interface, and import `report_kit`
2	**The keyword strategy** — three groups of terms: the core concept, the resistance *mechanisms*, and an explicit bacterial context. Generic "drug resistance" and bare acronyms are excluded on purpose
3	**Classification filters (IPC / CPC)**, one code list serving both
4	**Combine**: intersect the keyword families with the classification families
5	**One representative publication per family**, 2000 onwards, preferring **EP > WO > US**
6	**Export** `dataset.xlsx` — the single shared corpus notebooks 2 and 3 read
7	First chart — how the field has grown, by earliest filing year
8	Second chart — the twenty leading IPC classes

> **Trap — Watch the spacing in IPC symbols.** An 8-character IPC symbol is a 4-character sub-class, then padding spaces, then the main-group number: `A61K  31` is **two** spaces, `C12Q   1` is **three**. Get the padding wrong and the filter matches nothing — silently, with no error.
>
> **Trainer.** Step 5 is a decision worth naming: *one representative publication per family*, preferring EP then WO then US. That is a choice, and a different preference order would give a different citable number for the same invention. It is the module 4 lesson again — the defensible part is that it was written down.

Step 2 — The core landscape battery (10 min) · `2_core_landscape_analyses.ipynb`

Ten charts, all reading the same corpus. Do not walk through all ten — walk through the **structure**, then let the room read.

The question	The charts
Where?	filings by international/regional authority · national filing trends · national vs international filing strategy
When?	WO (PCT) vs EP over time · innovation waves by technology area
How far?	family size & global reach
Who?	top applicants by families · applicants by institutional sector · grant rate by top applicant · most influential organisations (forward citations)

> **Trainer.** The sector chart and the grant-rate chart are the two a PATLIB audience reacts to, because they separate universities from companies and separate *filing a lot* from *getting granted*. If time is short, spend it on those two and let the rest be read.
>
> **Trap.** The forward-citation analysis runs against **BigQuery**, not the TIP PATSTAT client — a citation self-join is the honest tool for that question and BigQuery is where it is affordable. That cell therefore needs separate credentials and will not run for a participant who only has TIP. Its output ships with the notebook.

Step 3 — One advanced analysis (4 min) · `3_advanced_analyses.ipynb`

A **technology co-occurrence network**: which IPC fields appear together on the same families. Five steps — load the corpus, map IPC codes to technology fields, keep the codes with at least 50 families, self-join to find co-occurrences, draw the network above an edge threshold.

This is the one chart in the report that answers a question the others cannot: **how the field is connected**, rather than how big it is or who is in it.

Step 4 — Assemble the report (4 min) · `4_assemble_report.ipynb`

Four steps, and this is where the pipeline pays off.

Every analysis in notebooks 1–3 ended with a `report_kit.record(...)` call that showed the figure inline *and* saved two things: the figure as an inline HTML fragment, and the data behind it. Notebook 4 collects every contribution from the output manifests, orders them, and stitches them into:

- **one self-contained HTML report** — a single embedded copy of `plotly.js`, **no iframes**, so it renders inside TIP and works with no internet at all. It has two view modes switched by a header button: **Paged** (one chart at a time, step bar, Previous/Next, ←/→ keys) and **One page** (everything stacked).
- **one data workbook** — one sheet per chart, so a client can check any number.

It is opened with the course's shared `open_html()` helper.

> **Trap — Never open a TIP-generated report with `IFrame`.** TIP's content-security policy sandboxes iframes and disables their JavaScript, so an interactive report renders as a blank or dead box. `open_html()` goes through jupyter-server-proxy and works. This applies to modules 6, 7 and 8 alike.

Step	What you do	What you see	Time
1	Build the corpus	One `dataset.xlsx`, and two charts	10 min
2	Run the core battery	Ten charts: where, when, how far, who	10 min
3	Run the network analysis	How the technology fields connect	4 min
4	Assemble	One HTML report + one workbook	4 min

Phase 3 · Learning outcome (≈ 10 min)

What now exists

- **One self-contained HTML landscape report** — **13 charts** (2 from notebook 1, 10 from notebook 2, 1 from notebook 3) — that opens on any machine with a browser and no internet.
- **One workbook** carrying the data behind every chart.
- **A written search strategy** — including its exclusions — that you can hand to anyone who disputes the corpus.

The third item is what makes the first two worth anything.

Self-check

1. **Why is `keywords AND (IPC OR CPC)` better than either half alone?** *(Keywords give recall but no precision; classification gives precision but misses anything classified oddly. The intersection is defensible in both directions.)*
2. **You are asked why MRSA is not in your search terms.** Your answer? *(Bare acronyms are noisy; precision comes from requiring a classification match. Note that this is an argued choice, not an oversight.)*
3. **Your IPC filter returns zero families and throws no error.** *(Check the padding: `A61K  31` is two spaces, `C12Q   1` is three.)*
4. **A client opens your report on a laptop with no internet. Does it work?** *(Yes — the plotly library is embedded and there are no external requests. That was a design requirement, not a convenience.)*

Transfer to your own work

Take the pipeline and change **only notebook 1**. Pick a technology your region actually cares about, write the three keyword groups and the classification list, and run notebooks 2–4 unchanged.

That is the real lesson of the module: the analyses are generic; **the corpus definition is the part that is yours**, and it is the part you have to be able to defend.

> **Trainer.** Close by opening the shipped report in Paged mode and walking the last three slides of it. The participants have just seen how each chart is made; seeing them assembled as a client-facing document is what makes the pipeline feel worth building.

Where this leads

Next	Why
Module 8 — IPScore	Module 6 describes a whole field. Module 8 asks what **one patent** in it is worth — and how much of that answer is evidence. The worked example in module 8 is drawn from *this* corpus.
Module 7 — IPScore (Riccardo's tools)	The companion valuation module, outside this course block.

Notes for the next revision

- The rebuild's `README.md` states that notebook 3 holds the technology network only, and that temporal, citation, t-SNE, SDG and triadic analyses are "the next phase". The repository-level `CLAUDE.md` describes module 6 as including t-SNE clusters and triadic families — that describes the older `1_antibiotic_resistance/` folder. **Align the two descriptions** before the workshop so nobody promises a chart that is not in the report.
- The forward-citation cell's BigQuery dependency should be stated in the notebook header, not only in the cell comment — a participant who tries to re-run notebook 2 on TIP alone will hit it.